J11: THE GRID OF THE TORUS WORD AND THE REFLECTED TORUS GRID

Abstract. Let $a \geq 2$ and $q \geq 1$. Write $\delta = \sigma_0\sigma_1\cdots\sigma_{a-2}$ for the Coxeter word on a strands. We prove that the selected grid diagram of the positive word δ^q and the row reflection of TauCeti's grid `torusLink(a-1)(q-1)` are related by grid moves. The case $a = S$, $q = S+1$ gives

`reflectedTorusGrid(S-1)`

$\sim$ `wordGrid(fullTwistWordS++coxeterWordS)`.

The proof replaces δ^q by the conjugate word β^q, where $\beta = \sigma_{a-2}\cdots\sigma_1\sigma_0$, and absorbs the factors β one at a time into a growing torus grid by explicit commutations and destabilizations. A preliminary computation of Jones polynomials and linking numbers confirms that no chirality obstruction occurs.

1. Statement, conventions and sources

1.1. **Grid diagrams.** We use the vendored TauCeti conventions. A grid diagram of size n is a pair (O, X) of permutations of $\{0, \ldots, n-1\}$, read as maps from columns to rows, with $O(c) \neq X(c)$ for every column c (`TauCeti.GridDiagram`). For $x \neq y$ in $\{0, \ldots, n-1\}$ the open cyclic interval $(x, y)^\circ$ is $\{z : x < z < y\}$ if $x < y$ and $\{z : z > x\} \cup \{z : z < y\}$ if $x > y$ (`Grid.cIoo`). Two pairs $\{x_0, x_1\}$, $\{y_0, y_1\}$ are *non-interleaving* if $x_0 \in (y_0, y_1)^\circ \Leftrightarrow x_1 \in (y_0, y_1)^\circ$ and $y_0 \in (x_0, x_1)^\circ \Leftrightarrow y_1 \in (x_0, x_1)^\circ$ (`Grid.Noninterleaving`).

The elementary moves are those of `GridDiagram.IsMove` in `External/TauCeti/KnotTheory/Grid/Move/Basic.lean`:

(1) a cyclic permutation of all rows or of all columns (`relabelRows(finRotaten)`, `relabelColumns(finRotaten)`);
(2) the exchange of two cyclically consecutive columns c, $c{+}1$ whose segments $\{O(c), X(c)\}$ and $\{O(c+1), X(c+1)\}$ are non-interleaving, and the analogous exchange of two consecutive rows, where the segment of a row is the pair of columns of its two markings (`movesTo_swapColumns`, `movesTo_swapRows`);
(3) a stabilization or destabilization of one of the eight types (`IsStabilization`, `IsDestabilization`).

We write $G \sim G'$ for the relation `MovesTo`, the equivalence relation generated by these moves. It relates diagrams of different sizes.

For destabilizations we use the following description, which is the content of `isXStabilization_eraseOCorner` in `GridMoveRecognition.lean` and of its mirror image obtained by exchanging O and X (`isDestabilization_swapMarkings`). Suppose that columns c, d are cyclically consecutive and rows r, s are cyclically consecutive, and that the corners (c, r), (d, r), (c, s) carry the markings O, X, X while (d, s) carries no O. Deleting column c and row r, and moving the X-marking of column d to row s, is an X-destabilization. Exchanging the letters O and X in this sentence gives an O-destabilization.

1.2. **The statement.** The Lean definitions involved are the following.

- `coxeterWorda=List.finRange(a-1)`, the word δ; `torusWordaq` $= \delta^q$, and `fullTwistWorda` $= \delta^a$ (`PositiveWordNormalForms.lean`). By `torusWord_succ`, $\delta^{S+1} = \delta^S\delta$, so `torusWordS(S+1)=fullTwistWordS++coxeterWordS`.

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- `wordGridw`, the grid field of the selected data of a word w. It is an admissible grid of w in the sense of Section 3 (`wordGrid_isAdmissible` in `PositiveWordGridAdmissible.lean`).
- `GridDiagram.torusLinkpq`, of size $N = (p+1)+(q+1)$, with $O(c) = c$ and $X(c) = c+q+1$ modulo N (`torusLink_O_apply`, `torusLink_X_apply`).
- `reflectedDiagramG=G.relabelRowsFin.revPerm`, so the reflected diagram has $O(c) = N - 1 - O_G(c)$ and $X(c) = N - 1 - X_G(c)$; and `reflectedTorusGridn=reflectedDiagram(torusLinkn(n+1))`.

Theorem 1.1. *Let $a \geq 2$ and $q \geq 1$. Then*

wordGrid(torusWordaq) ∼ reflectedDiagram(torusLink(a-1)(q-1)).

In particular, for $S \geq 2$,

reflectedTorusGrid(S-1) ∼ wordGrid(fullTwistWordS++coxeterWordS).

The second assertion is the case $a = S$, $q = S + 1$ of the first, because `torusLink(S-1)S` is the diagram reflected in `reflectedTorusGrid(S-1)`.

1.3. **Sources.** The proof is self-contained apart from the cited Lean declarations. The admissible grids of a word are described in chapter J10 of `docs/route-n/written-proofs-J05-J06-J07-J21-J10-J15-J14.tex` (section *The row-separated word grids*) and in `PositiveWordGridAdmissible.lean`. The statement that any two admissible grids of one word are related by row commutations is `IsAdmissibleWordGrid.movesTo` there. The conjugacy of Coxeter words under far commutations and rotations is `farRotationEquiv_coxeterWord` of `PositiveWordCoxeterConjugacy.lean` (job J08). The vendored TauCeti file on torus grids cites Ozsváth–Stipsicz–Szabó, *Grid homology for knots and links*, Chapter 3, which was not consulted. Nothing below depends on it.

2. A chirality check

The audit asks for a check that the two diagrams of Theorem 1.1 have the same chirality. We record a computation. It is evidence only, and the proof in later sections does not use it.

A grid diagram determines a planar diagram once one fixes which segment passes over the other at a crossing. Any fixed choice gives a combinatorial invariant of the relation $\sim$ only up to mirror image, but it allows us to compare two diagrams. With vertical segments over horizontal ones and the orientation from X to O in each column, we computed for both diagrams the Kauffman bracket polynomial normalized by the writhe, and, for links, the pairwise linking numbers. The computation (a Python script in the scratch directory of this job) gave equal values for `wordGrid`(δ^q) and `reflectedDiagram(torusLink(a-1)(q-1))` for $(a, q) \in \{(2,2),(2,3),(2,4),(3,3),(2,5),(3,4),(3,2),(4,5)\}$, and different values for the unreflected diagram `torusLink(a-1)(q-1)`. All linking numbers of the word grids are positive. For $(a, q) = (2, 3)$ the normalized bracket is $A^{-4} + A^{-12} - A^{-16}$ for the word grid and for the reflected torus grid. Thus the reflection in the statement is the expected one. A greedy search for destabilizations and commutations also reduced each of these word grids to a cyclic permutation of the reflected torus grid. The explicit sequence in Section 7 was found by analysing such reductions, and it was checked by machine for $2 \leq a \leq 6$, $1 \leq k \leq 5$ and for the words $v = \beta^j$, $0 \leq j \leq 2$. These checks are not part of the proof.

3. Labeled grid diagrams and elementary lemmas

Definition 3.1. A *labeled grid diagram* consists of a sequence $C = (c_0, \ldots, c_{n-1})$ of distinct column labels, a sequence $R = (r_0, \ldots, r_{n-1})$ of distinct row labels, and two maps O, X from $\{c_0, \ldots, c_{n-1}\}$ to $\{r_0, \ldots, r_{n-1}\}$ which are bijections and satisfy $O(c) \neq X(c)$ for every column label c. Its *position*

maps are $\mathrm{pos}(c_i) = i$ and $\mathrm{pos}(r_j) = j$. The associated grid diagram has $O'(i) = \mathrm{pos}(O(c_i))$ and $X'(i) = \mathrm{pos}(X(c_i))$.

The associated grid diagram is a grid diagram because O, X are bijections and $O(c) \neq X(c)$. Relabeling the labels by bijections that preserve both sequences does not change it. We say that a row label r has its markings in the columns $O^{-1}(r)$ and $X^{-1}(r)$.

Moves are described on labeled diagrams. Exchanging two consecutive entries of C (or of R) produces a labeled diagram whose associated diagram is obtained by `swapColumns` (or `swapRows`) of the two positions. The exchange is a grid move when the two segments are non-interleaving. Moving the first entry of C to the end is a cyclic permutation of columns. A destabilization deletes one column label and one row label and changes one value of O or X.

Lemma 3.2. *An exchange of two rows does not change the position of any column, so it does not change the segments of columns. An exchange of two columns does not change the segments of rows except the two exchanged columns. Consequently, if a row r is non-interleaving with each of the rows $s_1, \dots, s_\ell$ that are consecutive to it in this order, then r can be moved past $s_1, \dots, s_\ell$ by ℓ row exchanges, each of which is a grid move. The same holds for columns.*

Proof. The first two sentences are immediate from the definitions. The segment of the row r consists of the positions of the two columns containing its markings, and the segments of $s_1, \dots, s_\ell$ likewise. Row exchanges do not change these positions, so the non-interleaving hypotheses hold at every step. □

Lemma 3.3 (short segments)**.** *Let c be a column whose two markings lie in rows with consecutive positions. Then c is non-interleaving with every column d that has no marking in these two rows. The same holds for a row whose two markings lie in columns with consecutive positions, and every row with no marking in these two columns.*

Proof. Let the rows of c have positions $x, x+1$. The cyclic interval between them, in either direction, contains no position or all positions other than $x, x+1$. The markings of d avoid $x, x+1$, so they lie on one side. In the other direction, x and $x+1$ are both inside or both outside the cyclic interval between the rows of d, because no position lies strictly between them. This is `columnsNoninterleaving_of_adjacent_rows` and `rowsNoninterleaving_of_adjacent_columns` of `GridMoveRecognition.lean`, and J10, Lemma *commutations of a short row or column.* □

Lemma 3.4 (separated rows)**.** *Let W be a set of column positions forming an interval $\{p, p+1, \dots, p'\}$ of integers. If the markings of a row r lie in columns with positions in W and the markings of a row s lie in columns with positions outside W, then r and s are non-interleaving.*

Proof. Let $x_0, x_1 \in W$ be the positions of the columns of r and $y_0, y_1 \notin W$ those of s. The set $(x_0, x_1)^\circ$ is either contained in W or contains every position outside W, so y_0 and y_1 lie on the same side of it. The set $(y_0, y_1)^\circ$ either contains W or is disjoint from W, because W is an interval of positions avoiding y_0, y_1. So x_0 and x_1 lie on the same side. □

4. The torus grids

Definition 4.1. Let $a \geq 2$ and $k \geq 1$, and put $N = a + k$. The grid diagram $R_{a,k}$ of size N has

$$O(c) = -c \bmod N, \qquad X(c) = a - c \bmod N.$$

Since $0 < a < N$, the residues $-c$ and $a - c$ differ, and both maps are permutations. So $R_{a,k}$ is a grid diagram.

Lemma 4.2. *For $a \geq 2$ and $q \geq 1$, put $N = a + q$. Then $R_{a,q}$ is obtained from*

`reflectedDiagram(torusLink(a-1)(q-1))`

by the cyclic permutation of rows `relabelRows(finRotateN)`*. In particular the two diagrams are related by one grid move.*

Proof. The diagram `torusLink(a-1)(q-1)` has size $(a-1+1)+(q-1+1) = N$, with $O(c) = c$ and $X(c) = c+q \bmod N$. Its reflection has $O(c) = N-1-c$ and $X(c) = N-1-(c+q \bmod N)$. Adding one modulo N to every row gives $O(c) \equiv -c$ and $X(c) \equiv -c-q \equiv a-c$, because $N = a+q$. □

In $R_{a,k}$, the row t has its O-marking in column $-t$ and its X-marking in column $a-t$, modulo N. We call the rows $1, \dots, a$ the *window rows*. These are exactly the rows whose O-marking lies in one of the last a columns $N-a, \dots, N-1$. The window row $r+1$, for $0 \le r \le a-1$, has its O-marking in column $N-1-r$ and its X-marking in column $a-1-r$. We call r its *rank*.

5. Hybrid grids

Fix $a \ge 2$. A positive word v on a strands *covers all ranks* if for every $r \in \{0, \dots, a-1\}$ some letter of v involves the rank r, that is, equals σ_{r-1} or σ_r. Every word containing all generators covers all ranks, by `positiveRectilinearWord_all_rows_occur`. We recall the arcs of the admissible grids of a word v of length L (J10, section *The row-separated word grids*). The corners are the pairs (h, c) with $0 \le h < L$ and $c \in \{0, 1, 2\}$, in lexicographic order. Every corner is the start of one horizontal arc, the *arc leaving* the corner. For the letter σ_s in position h, the arcs leaving $(h,0), (h,1), (h,2)$ have the heights $s+\frac{1}{2}$, s, $s+1$. The arc leaving $(h,0)$ ends at $(h,2)$. An arc of integer height r leaving an occurrence of the rank r ends at the incoming corner of the next occurrence of r, where the incoming corner of an occurrence is $(h,0)$ if r is the lower rank of the letter and $(h,1)$ if it is the upper rank, and the outgoing corner is $(h,1)$, respectively $(h,2)$.

Definition 5.1. Let $k \ge 1$, $N = a+k$, and let v be a positive word on a strands which is empty or covers all ranks. The labeled grid diagram $H_{a,k}(v)$ has the following data.

1. Columns: the labels $\rho_0, \dots, \rho_{N-1}$, followed by the corners of v in lexicographic order.
2. Rows: the labels ϱ_t for $t \in \{0\} \cup \{a+1, \dots, N-1\}$; for each rank r a label ω_r if v is empty and a label ι_r if v is nonempty; and the arcs of v.
3. The O-marking of ρ_c is ϱ_t with $t = -c \bmod N$ if this residue is not a window row, and ω_r, respectively ι_r, if $-c \equiv r+1$ with $0 \le r \le a-1$.
4. The X-marking of ρ_c is ϱ_t with $t = a-c \bmod N$ if this residue is not a window row. If $a-c \equiv r+1$ with $0 \le r \le a-1$, it is ω_r when v is empty and the arc of v leaving the outgoing corner of the last occurrence of r in v when v is nonempty.
5. The O-marking of a corner of v is the arc leaving it. The X-marking of the incoming corner of the first occurrence of a rank r in v is ι_r. The X-marking of every other corner is the arc of v ending there.
6. Row order: first ϱ_0; then the *band*, consisting of the labels ω_r, ι_r of height r and the arcs of v with their heights, ordered by height with an arbitrary order among equal heights; then $\varrho_{a+1}, \dots, \varrho_{N-1}$.

So $H_{a,k}(v)$ is $R_{a,k}$ with the word grid of v inserted at the right end: the window row of rank r is cut into the row ι_r, which ends at the first occurrence of r in v, and the arc of v leaving the last occurrence of r, which ends in column ρ_{a-1-r}.

Lemma 5.2. *$H_{a,k}(v)$ is a labeled grid diagram. The diagram $H_{a,k}(\varnothing)$ is $R_{a,k}$. Any two choices of the order among equal heights in the band give diagrams related by row exchanges, each of which is a grid move.*

Proof. Each row label is the O-marking of exactly one column: ϱ_t of ρ_{-t}, ω_r and ι_r of ρ_{N-1-r}, and an arc of v of its starting corner. Each row label is the X-marking of exactly one column: ϱ_t of ρ_{a-t}, ω_r of ρ_{a-1-r}, ι_r of the first incoming corner of r, the arc leaving the last occurrence of r

of ρ_{a-1-r}, and every other arc of v of its end corner. For the columns ρ_c the two markings differ because $-c \not\equiv a-c$, and an O-label of the form ι_r is never an X-label of an R-column. For a corner of v, the arc leaving it and the label at its X-marking are different, because the matching of v has no fixed point (`positiveRectilinearWordMatching_ne_self`) and because ι_r is not an arc of v.

If v is empty, the rows are $\varrho_0, \omega_0, \ldots, \omega_{a-1}, \varrho_{a+1}, \ldots, \varrho_{N-1}$ in this order, and ω_r occupies the window row $r+1$. So the associated diagram is $R_{a,k}$.

For the last assertion it suffices, by Lemma 3.2 and the argument of J10, Lemma *same-height rows do not interleave*, to show that two rows of the band of equal height are non-interleaving. At a half-integer height both rows are middle arcs of different letters of v, whose column intervals are disjoint. At an integer height r the rows are: ι_r, with columns ρ_{N-1-r} and the first incoming corner f of r; the arc λ leaving the last outgoing corner ℓ of r, with columns ℓ and ρ_{a-1-r}; and the arcs of v joining consecutive occurrences of r, whose column intervals lie in the closed interval from f to ℓ and are pairwise disjoint (J10, loc. cit.). In the linear order of positions, $\rho_{a-1-r} < \rho_{N-1-r} < f < \ell$. Hence the interval of ι_r lies inside the interval of λ, the arcs between consecutive occurrences lie inside the interval of λ and are disjoint from the interval of ι_r. All endpoints are distinct, because the incoming and outgoing corners of an occurrence are different. So every pair is nested or disjoint, hence non-interleaving. □

6. The descending Coxeter block

Let $\beta = \sigma_{a-2}\sigma_{a-3}\cdots\sigma_0$. Its letter in position t is σ_{a-2-t}, for $0 \le t \le a-2$. Every rank occurs in β: the rank $a-1$ only in position 0, the rank 0 only in position $a-2$, and a rank r with $1 \le r \le a-2$ in positions $a-2-r$ (lower rank) and $a-1-r$ (upper rank). We use the following names for the corners of β and the arcs leaving them.

$D_t = (t,1)$, $0 \le t \le a-2$	the descending columns
$A_r = (a-2-r,0)$, $B_r = (a-2-r,2)$, $0 \le r \le a-2$	the ascending columns of rank r
μ_r, the arc leaving A_r	height $r+\frac{1}{2}$, from A_r to B_r
e_r, the arc leaving D_{a-2-r}, $1 \le r \le a-2$	height r, from D_{a-2-r} to D_{a-1-r}
o_0, the arc leaving D_{a-2}; o_r, the arc leaving B_{r-1}, $1 \le r \le a-1$	height r, the last outgoing arc of rank r

In a diagram containing β as a block of consecutive columns, ι_r denotes the row of the X-marking of the first incoming corner of r, namely A_r for $r \le a-2$ and D_0 for $r = a-1$. Thus the markings of the block columns are

	D_0	D_t $(1 \le t \le a-3)$	D_{a-2} $(a \ge 3)$	A_r
O	e_{a-2}	e_{a-2-t}	o_0	μ_r
X	ι_{a-1}	e_{a-1-t}	e_1	ι_r

	B_r
O	o_{r+1}
X	μ_r

For $a = 2$ the only descending column is D_0, with $O = o_0$ and $X = \iota_1$. The column order of the block is $A_{a-2}, D_0, B_{a-2}, A_{a-3}, D_1, B_{a-3}, \ldots, A_0, D_{a-2}, B_0$.

Definition 6.1. A labeled grid diagram G is in *block position* if it contains these $3(a-1)$ column labels as consecutive columns in the order above, with the markings above, where the labels $\iota_0, \ldots, \iota_{a-1}, o_0, \ldots, o_{a-1}, e_1, \ldots, e_{a-2}, \mu_0, \ldots, \mu_{a-2}$ are pairwise distinct, except that $\iota_r = o_r$ is allowed, and if the row order satisfies:

(B1) for every r, all rows among ι_r, o_r, e_r lie below μ_r, and μ_r lies below all rows among $\iota_{r+1}, o_{r+1}, e_{r+1}$;

(B2) e_r lies below ι_r for $1 \le r \le a-2$, and o_0 lies below ι_0 unless $o_0 = \iota_0$.

The next lemma performs the two first stages of the reduction. Its moves only use the rows and columns of the block, together with Lemma 3.3, so the rest of the diagram is arbitrary.

Lemma 6.2 (block reduction)**.** *Let G be in block position, and assume $\iota_r \neq o_r$ for every r. Then $G \sim G'$, where G' is obtained from G as follows: the block columns are replaced by the a consecutive columns*

$$V,\ B_{a-2},\ B_{a-3},\ \ldots,\ B_0,$$

with $O(V) = o_0$, $X(V) = \iota_{a-1}$, $O(B_r) = o_{r+1}$ and $X(B_r) = \iota_r$; the rows $e_1, \ldots, e_{a-2}, \mu_0, \ldots, \mu_{a-2}$ are deleted; and the relative order of all other rows, and of all columns outside the block, is unchanged.

Proof. Stage 1: merging the descending columns. We show by induction on $t \in \{0, \ldots, a-2\}$ that G is related to the diagram G_t in which the block columns are

$$A_{a-2},\ M_t,\ B_{a-2}, A_{a-3}, B_{a-3}, \ldots, A_{a-2-t}, B_{a-2-t},\ A_{a-3-t}, D_{t+1}, B_{a-3-t}, \ldots, A_0, D_{a-2}, B_0,$$

where M_t is the label D_t with $X(M_t) = \iota_{a-1}$ and the other markings unchanged, and the rows $e_{a-1-t}, \ldots, e_{a-2}$ (an empty list for $t = 0$) are deleted. For $t = 0$ this is G. The case $t = a-2$ gives the order $A_{a-2}, V, B_{a-2}, A_{a-3}, B_{a-3}, \ldots, A_0, B_0$ with $V = M_{a-2}$, $O(V) = o_0$, $X(V) = \iota_{a-1}$.

Let $t \leq a-3$. The column D_{t+1} has X-marking e_{a-2-t} and O-marking e_{a-3-t} if $t \leq a-4$, and o_0 if $t = a-3$. Its rows have heights $a-2-t$ and $a-3-t$. We move D_{t+1} to the left until it is immediately right of M_t. It crosses the columns A_{a-3-t} and B_s, A_s for $a-2-t \leq s \leq a-2$, $s \neq a-2$ for A_s. By Lemma 3.2 it suffices to check that each of them is non-interleaving with D_{t+1}, and this depends only on the order of their rows.

- A_{a-3-t} has rows ι_{a-3-t} and μ_{a-3-t}. By (B2) the O-row of D_{t+1} lies below ι_{a-3-t}. By (B1) μ_{a-3-t} lies below e_{a-2-t}. So the segment of A_{a-3-t} lies strictly inside the segment of D_{t+1} in the linear order of rows. Nested segments with distinct endpoints are non-interleaving.
- A_s with $a-2-t \leq s \leq a-3$ has rows ι_s, μ_s of heights $s, s+\frac{1}{2}$. For $s = a-2-t$, (B2) puts e_{a-2-t} below ι_s, and for larger s the heights separate. So the segments are disjoint.
- B_s with $s \geq a-2-t$ has rows μ_s, o_{s+1} of heights $s+\frac{1}{2}, s+1$, all above the rows of D_{t+1} by (B1). The segments are disjoint.

Now M_t and D_{t+1} are consecutive columns and the row e_{a-2-t} has its O-marking in M_t and its X-marking in D_{t+1}. Move this row up until it is immediately below ι_{a-1}. The only rows with markings in M_t or D_{t+1} are ι_{a-1}, e_{a-2-t} and the O-row of D_{t+1}, which lies below e_{a-2-t}. So every row crossed has no marking in these two columns, and Lemma 3.3 applies at every step. Now the corners (M_t, e_{a-2-t}), (D_{t+1}, e_{a-2-t}), (M_t, ι_{a-1}) carry O, X, X, and (D_{t+1}, ι_{a-1}) carries no O, because the O-row of D_{t+1} has height at most $a-3-t < a-1$. The X-destabilization deletes M_t and e_{a-2-t} and gives $X(D_{t+1}) = \iota_{a-1}$. Renaming D_{t+1} as M_{t+1} gives G_{t+1}. The rows other than e_{a-2-t} keep their relative order.

Stage 2: removing the middle arcs. Treat $r = a-2, a-3, \ldots, 0$ in turn. If $r = a-2$, first move A_{a-2} to the right past V. Its segment, with rows ι_{a-2} and μ_{a-2}, lies strictly inside the segment of V, with rows o_0 and ι_{a-1}: by (B1) and (B2), o_0 lies below ι_{a-2} (for $a = 2$ this is $o_0 < \iota_0$) and μ_{a-2} lies below ι_{a-1}. For $r \leq a-3$ the columns A_r and B_r are already consecutive. Now the row μ_r has its markings in the consecutive columns A_r, B_r. Move it down until it is immediately above ι_r. The rows with markings in A_r or B_r are ι_r, μ_r and o_{r+1}, and o_{r+1} lies above μ_r, so Lemma 3.3 applies at every step. The corners (A_r, μ_r), (B_r, μ_r), (A_r, ι_r) carry O, X, X and (B_r, ι_r) carries no O, because $O(B_r) = o_{r+1}$. The X-destabilization deletes A_r and μ_r and gives $X(B_r) = \iota_r$.

After both stages the block columns are $V, B_{a-2}, \ldots, B_0$ with the stated markings. Each step deleted one of the listed rows or exchanged rows only to bring a row that is later deleted next to another row, so the relative order of the remaining rows is unchanged. □

7. Absorption of one block

Lemma 7.1 (absorption)**.** *Let $a \geq 2$, $k \geq 1$, and let v be a positive word on a strands which is empty or covers all ranks. Then $H_{a,k}(\beta v) \sim H_{a,k+1}(v)$.*

Proof. Put $N = a + k$. The word βv covers all ranks, so $H = H_{a,k}(\beta v)$ has the rows ι_r. In βv the first occurrence of every rank lies in β, so ι_r is the X-marking of the first incoming corner of r in the block β, and the notation of Section 6 applies. The arc o_r leaving the last occurrence of r in β ends at the first occurrence of r in v if v is nonempty, and in the column ρ_{a-1-r} if v is empty. By Lemma 5.2 we may choose the order among rows of equal height in the band. We choose it so that, for every r, the row o_r is the lowest and ι_r the highest row of height r. Then (B1) holds because the band is ordered by height, and (B2) holds by this choice. Also $\iota_r \neq o_r$, since ι_r is not an arc. So H is in block position and Lemma 6.2 gives $H \sim H'$, where the block is replaced by $V, B_{a-2}, \dots, B_0$.

Stage 3. For $r = 0, 1, \dots, a-2$ in turn we do the following. The column ρ_{N-1-r} has O-marking ι_r, and $X(B_r) = \iota_r$, $O(B_r) = o_{r+1}$.

(1) Move the row ι_r up until it is immediately below o_{r+1}. In the current diagram, the markings of ι_r lie in ρ_{N-1-r} and B_r. Since ι_r is the highest row of height r and o_{r+1} is the lowest row of height $r+1$, and since the middle arcs of the block were deleted in Lemma 6.2, every row strictly between them is a middle arc of v of height $r + \frac{1}{2}$. Such an arc has both markings in columns of v. The columns $\rho_0, \dots, \rho_{N-1}$ followed by the block columns form an interval of positions containing both markings of ι_r, and the columns of v lie outside it. By Lemma 3.4 and Lemma 3.2, all these exchanges are grid moves.

(2) Move the column B_r to the left until it is immediately right of ρ_{N-1-r}. The rows of B_r now have consecutive positions. The columns crossed are V, $B_{a-2}, \dots, B_{r+1}$ and $\rho_{N-r}, \dots, \rho_{N-1}$. Their markings are: o_0, ι_{a-1} for V; o_{s+1}, ι_s for B_s; and for ρ_{N-1-s} with $s < r$, the O-marking o_{s+1} (it was changed in the step for s) and an X-marking. This X-marking is the label of the residue $a - (N-1-s) \equiv s+1-k$ modulo N: a row ϱ_t if $s < k$, and otherwise the window row of rank $u = s-k$, namely an arc of v, or the row o_u if v is empty. Since $u \leq s < r$, none of these markings lies in the rows ι_r, o_{r+1}. So Lemma 3.3 applies at every step.

(3) The corners (ρ_{N-1-r}, ι_r), (B_r, ι_r), (B_r, o_{r+1}) carry O, X, O, and the corner (ρ_{N-1-r}, o_{r+1}) carries no X-marking, by the description of $X(\rho_{N-1-r})$ just given with $s = r$ (for v empty and $r \geq k$, $X(\rho_{N-1-r}) = o_{r-k} \neq o_{r+1}$). The O-destabilization deletes B_r and ι_r and gives $O(\rho_{N-1-r}) = o_{r+1}$.

Identification. After stage 3 the columns are $\rho_0, \dots, \rho_{N-1}, V$, followed by the corners of v. The markings are those of $H_{a,k}(\beta v)$ for all columns ρ_c except $O(\rho_{N-1-r}) = o_{r+1}$ for $0 \leq r \leq a-2$; moreover $O(V) = o_0$, $X(V) = \iota_{a-1}$, and the corners of v keep their markings, with X-marking o_r at the first incoming corner of r in v. Rename the labels by

$$\rho_c \mapsto \rho'_c \ (c < N), \quad V \mapsto \rho'_N, \quad \varrho_0 \mapsto \varrho'_0, \quad \varrho_t \mapsto \varrho'_{t+1} \ (t \geq a+1), \quad \iota_{a-1} \mapsto \varrho'_{a+1},$$

$$o_r \mapsto \iota'_r \text{ if } v \text{ is nonempty}, \qquad o_r \mapsto \omega'_r \text{ if } v \text{ is empty},$$

and keep the arcs and corners of v. We compare with $H_{a,k+1}(v)$, of size $N+1$, column by column, using residues modulo $N+1$ for $H_{a,k+1}(v)$ and modulo N for $H_{a,k}(\beta v)$.

- O-markings. In $H_{a,k}(\beta v)$ after stage 3: $O(\rho_0) = \varrho_0$; $O(\rho_c) = \varrho_{N-c}$ for $1 \leq c \leq k-1$; $O(\rho_k) = \iota_{a-1}$; $O(\rho_c) = o_{N-c}$ for $k+1 \leq c \leq N-1$; and $O(V) = o_0$. In $H_{a,k+1}(v)$: $O(\rho'_0) = \varrho'_0$; $O(\rho'_c) = \varrho'_{N+1-c}$ for $1 \leq c \leq k$, since then $N+1-c \geq a+1$; and $O(\rho'_c)$ is the row ι'_{N-c} (or ω'_{N-c}) for $k+1 \leq c \leq N$. Under the renaming these agree: for $c \leq k-1$ because $\varrho_{N-c} \mapsto \varrho'_{N-c+1}$, for $c = k$ because $\iota_{a-1} \mapsto \varrho'_{a+1} = \varrho'_{N+1-k}$, and for $k+1 \leq c \leq N$ because $o_{N-c} \mapsto \iota'_{N-c}$ (with $V = \rho'_N$ and $o_0 \mapsto \iota'_0$).
- X-markings of the columns ρ_c, $c \leq N-1$. For $0 \leq c \leq a-1$ both diagrams have the window row of rank $a-1-c$, which is the arc of v leaving its last occurrence of $a-1-c$ if v is nonempty, and $o_{a-1-c} \mapsto \omega'_{a-1-c}$ if v is empty. For $c = a$ both have the row of index 0,

that is $\varrho_0 \mapsto \varrho'_0$. For $a+1 \le c \le N-1$, the diagram $H_{a,k}(\beta v)$ has ϱ_{N+a-c} and $H_{a,k+1}(v)$ has $\varrho'_{N+1+a-c}$, which agree.
- $X(V) = \iota_{a-1} \mapsto \varrho'_{a+1}$, and in $H_{a,k+1}(v)$ the column ρ'_N has X-marking $a - N \equiv a+1 \pmod{N+1}$; since $k \ge 1$, $a+1 \le N$ is not a window row.
- Corners of v: the incoming corner of the first occurrence of r has X-marking o_r, which is ι'_r; the other markings are arcs of v in both diagrams. When v is empty there are no such columns.

The rows of the renamed diagram are ϱ'_0, then the band rows, then $\varrho'_{a+1} = \iota_{a-1}$ and $\varrho'_{a+2}, \dots, \varrho'_N$. Indeed ι_{a-1} was the highest row of height $a-1$ and nothing above it in the band was moved, and the row exchanges of stage 3 only moved rows ι_r inside the band before deleting them. The band rows are ordered by height, because each o_r kept its position, the deleted rows were the only ones moved, and the arcs of v never moved. So the renamed diagram is $H_{a,k+1}(v)$ with some order among equal heights, and Lemma 5.2 completes the proof. □

8. The first block

Lemma 8.1 (base case)**.** *Let $a \ge 2$ and let v be a nonempty positive word on a strands that covers all ranks. Then every admissible grid of βv is related to $H_{a,1}(v)$.*

Proof. By `IsAdmissibleWordGrid.movesTo` we may choose the admissible grid. View it as the labeled diagram whose columns are the corners of βv and whose rows are its arcs, ordered by height with the following choice among equal heights: o_r lowest and the arc ι_r highest among the arcs of height r. Here ι_r is the arc of v leaving the last occurrence of r in v; it ends at the first occurrence of r, which lies in β. Since v covers all ranks, $\iota_r \ne o_r$, and the diagram is in block position. Lemma 6.2 replaces the block by $V, B_{a-2}, \dots, B_0$.

Perform the O-stabilization of the O-marking of V that inserts a new column κ immediately right of V and a new row ν immediately below o_0: afterwards $O(V) = \nu$, $O(\kappa) = o_0$ and $X(\kappa) = \nu$. This is the TauCeti construction `stabilizeO` with the new column and the new row cyclically adjacent to the split column and to its O-row, so it is an elementary stabilization. Next move ν down to the bottom. Its markings lie in the consecutive columns V, κ, and the rows with markings in these columns are ν, o_0 and ι_{a-1}, both above ν. So Lemma 3.3 applies. Next move κ to the right past $B_{a-2}, \dots, B_0$. The segment of κ has rows ν and o_0, and the segment of B_r has rows ι_r and o_{r+1}, which lie above o_0 (for $r = 0$ because o_0 is the lowest row of height 0). The segments are disjoint, so these are grid moves.

The columns are now $V, B_{a-2}, \dots, B_0, \kappa$ followed by the corners of v, and the rows are ν followed by the arcs of βv that were not deleted, in an order by height. Rename $V \mapsto \rho_0$, $B_r \mapsto \rho_{a-1-r}$, $\kappa \mapsto \rho_a$, $\nu \mapsto \varrho_0$ and $o_r \mapsto \iota'_r$, keeping the arcs and corners of v. In $H_{a,1}(v)$, of size $a+1$, the column ρ_c has O-marking $-c \bmod (a+1)$ and X-marking $a-c$. Thus ρ_0 has $O = \varrho_0$ and X the window row of rank $a-1$, that is the arc of v leaving its last occurrence of $a-1$; this arc is $\iota_{a-1} = X(V)$. For $1 \le c \le a-1$, $\rho_c = B_{a-1-c}$ has O the window row $a+1-c$ of rank $a-c$, which is $\iota'_{a-c} = o_{a-c} = O(B_{a-1-c})$, and X the window row of rank $a-1-c$, which is the arc $\iota_{a-1-c} = X(B_{a-1-c})$. Finally $\rho_a = \kappa$ has $O = \iota'_0 = o_0$ and $X = \varrho_0 = \nu$. The corners of v have the markings of Definition 5.1, with X-marking $o_r = \iota'_r$ at the first incoming corner of r in v. The rows are ϱ_0 followed by the band, ordered by height. □

9. Proof of Theorem 1.1

9.1. Conjugacy of the words.

Lemma 9.1. *Let γ, γ' be positive words related by far commutations inside the word and by rotations $u\,i \mapsto i\,u$ (the relation `FarRotationEquiv`), and let $q \ge 1$. If γ is nonempty, then γ^q and γ'^q are related by `PositiveWordConj`.*

Proof. It suffices to treat one step. If $\gamma = u\,i\,j\,v \mapsto u\,j\,i\,v$ is a far commutation, then $\gamma^q \mapsto \gamma'^q$ is the composition of q far commutations, one in each factor. If $\gamma = u\,i$ and $\gamma' = i\,u$, then $\gamma^q = (u\,i)^q$, and the rotation moving the last letter i to the front gives $i\,(u\,i)^{q-1}u = (i\,u)^q$. Both are generators of `PositiveWordConj`. The words in each step are nonempty, so reflexivity is never needed. □

The word β is a permutation of $\delta =$ `List.finRange(a-1)`, so `farRotationEquiv_coxeterWord` gives $\beta \approx \delta$ in `FarRotationEquiv`. For $a \geq 2$ both words are nonempty, and Lemma 9.1 gives $\beta^q \sim \delta^q$ in `PositiveWordConj`. Every generator occurs in δ^q.

Lemma 9.2 (cyclic and far moves, J09)**.** *Let u, v be positive words such that v is obtained from u by one rotation or one far commutation. Then `wordGridu` $\sim$ `wordGridv`.*

Proof. This is proved in `docs/route-n/J09-word-grid-moves.tex`, Propositions on rotations and far commutations. For a rotation $u = u'\,i \mapsto i\,u'$, three cyclic permutations of columns move the three columns of the last letter to the front, and the result is an admissible grid of $i\,u'$. For a far commutation $u'\,i\,j\,v' \mapsto u'\,j\,i\,v'$ with $|i-j| \geq 2$, the rows of the columns of the letter i have heights in $[i, i+1]$ and those of the letter j in $[j, j+1]$; these intervals are disjoint, so the nine column exchanges that move the three columns of j before those of i are grid moves, and the result is an admissible grid of $u'\,j\,i\,v'$. In both cases `IsAdmissibleWordGrid.movesTo` relates the result to the selected grid. □

Applying Lemma 9.2 along a sequence of rotations and far commutations gives

$$\texttt{wordGrid}(\delta^q) \sim \texttt{wordGrid}(\beta^q). \tag{1}$$

9.2. **The case $q \geq 2$.** Let $q \geq 2$. The word β^{q-1} is nonempty and covers all ranks. Lemma 8.1 gives $\texttt{wordGrid}(\beta^q) \sim H_{a,1}(\beta^{q-1})$. For $1 \leq k \leq q-1$, Lemma 7.1 with $v = \beta^{q-1-k}$ gives $H_{a,k}(\beta^{q-k}) \sim H_{a,k+1}(\beta^{q-1-k})$. The word β^{q-1-k} is empty or covers all ranks. Composing, $\texttt{wordGrid}(\beta^q) \sim H_{a,q}(\varnothing) = R_{a,q}$ by Lemma 5.2. With (1) and Lemma 4.2 this proves the theorem for $q \geq 2$.

9.3. **The case $q = 1$.** For $q = 1$ the right side is the reflection of `torusLink(a-1)0`, which is related by Lemma 4.2 to $R_{a,1}$. In $R_{a,1}$, of size $a+1$, every column has $X(c) \equiv O(c) - 1$. We show that $R_{a,1}$ and the word grid of δ are both related to the diagram $U = (O, X) = ((0,1),(1,0))$ of size 2.

The diagram $R_{a,1}$. Let $n \geq 3$ and let G be a grid diagram of size n with $O(c) = -c$ and $X(c) = -c-1$ modulo n. The columns $c = n-2$ and $c+1 = n-1$ are consecutive; the rows 2 and 1 are consecutive; the corners $(n-2,2), (n-1,1), (n-2,1)$ carry O, O, X, since $O(n-2) = 2$, $O(n-1) = 1$, $X(n-2) = 1$, and $(n-1,2)$ carries no X because $X(n-1) = 0$. So this is an O-destabilization deleting column $n-2$ and row 1 and setting $O(n-1) = 2$. After renumbering, the remaining diagram has $O(c) = -c$, $X(c) = -c-1$ modulo $n-1$: for $c \leq n-3$ the rows $-c, -c-1$ are at least 2 or equal to 0, and they are renumbered by subtracting one when they exceed 1, which turns residues modulo n into residues modulo $n-1$; the last column $n-2$ has $O = 2-1 = 1 \equiv -(n-2)$ and $X = 0 \equiv -(n-2)-1$. By induction, $R_{a,1} \sim G_2$, where G_2 has $O = (0,1)$ and $X = (1,0)$, which is U.

The word grid of δ. Let $w = \delta$ on a strands, of length $a-1$. In an admissible grid, the rank 0 occurs only in letter 0, so the arc leaving $(0,1)$ is the only arc of height 0, it ends at $(0,0)$, and it is the lowest row. The arc leaving $(0,0)$ has height $\frac{1}{2}$ and is the second row. So column 0 has X in row 0 and O in row 1, column 1 has O in row 0, and $X(1)$ is the arc ending at the incoming upper corner of σ_0, of height 1, so $X(1) \neq 1$. The corners $(0,0), (0,1), (1,0)$ carry X, O, O, and $(1,1)$ carries no X: this is an O-destabilization deleting column 0 and row 0, with $O(1) = 1$. We show by induction on a that the word grid of δ on a strands is related to U; for $a = 2$ the word is σ_0, of size 3, and the destabilization just described gives a diagram of size 2, which is U or its cyclic row shift. For $a \geq 3$ we perform, after the destabilization above, two further destabilizations which

remove the remaining columns $(0,1)$ and $(0,2)$ of the first letter, and we verify that the result is an admissible grid of $\sigma_1 \cdots \sigma_{a-2}$ on the strands $1, \ldots, a-1$, which is the word grid of δ on $a-1$ strands after relabeling ranks.

Write the arcs of δ (for $q = 1$) as follows: μ_h leaving $(h, 0)$; for $1 \le r \le a-2$, the arc c_r leaving $(r-1, 2)$ and ending at $(r, 0)$, and the arc d_r leaving $(r, 1)$ and ending at $(r-1, 1)$; the arc d_0 leaving $(0, 1)$ and ending at $(0, 0)$; and the arc c_{a-1} leaving $(a-2, 2)$ and ending at $(a-2, 1)$. Choose the order among equal heights with c_r below d_r. The rows are then

$$d_0 < \mu_0 < c_1 < d_1 < \mu_1 < c_2 < d_2 < \cdots < \mu_{a-2} < c_{a-1}.$$

After the first destabilization (deleting column $(0,0)$ and row d_0, with $O((0,1)) = \mu_0$), the columns $(0,1)$ and $(0,2)$ are the first two columns, with markings $O = \mu_0, X = d_1$ and $O = c_1, X = \mu_0$. The rows μ_0 and c_1 are the two lowest rows. The corners $((0,2), c_1), ((0,2), \mu_0), ((0,1), \mu_0)$ carry O, X, O and $((0,1), c_1)$ carries no X, since $X((0,1)) = d_1 \neq c_1$. The O-destabilization deletes column $(0,2)$ and row μ_0 and gives $O((0,1)) = c_1$. Now column $(0,1)$ is first, with $O = c_1$ and $X = d_1$, the rows c_1, d_1 are the two lowest rows, and column $(1,0)$ is second, with $X = c_1$ and $O = \mu_1$. The corners $((0,1), c_1), ((0,1), d_1), ((1,0), c_1)$ carry O, X, X and $((1,0), d_1)$ carries no O. The X-destabilization deletes column $(0,1)$ and row c_1 and gives $X((1,0)) = d_1$. In the remaining diagram the arc d_1 leaves $(1,1)$ and ends at $(1,0)$, and it is the lowest row; all other markings are those of the letters $1, \ldots, a-2$. This is the admissible grid of $\sigma_1 \cdots \sigma_{a-2}$ in which d_1 plays the role of d_0, with the same order convention. By induction the word grid of δ is related to U or to its cyclic row shift, and U is related to its cyclic row shift by one cyclic permutation. Hence $\texttt{wordGrid}(\delta) \sim R_{a,1}$, and the theorem holds for $q = 1$.

Remark 9.3. The three destabilizations per letter in the case $q = 1$ were found by machine and agree with the sequence printed by the search: an O-destabilization, then an X-destabilization, at the first columns. The verification above is by hand.

10. Formalization

The case $q \ge 2$ of Theorem 1.1 is formalized without additional hypotheses in `TorusWordGridMoves.lean` (`wordGrid_torusWord_movesTo` and, for $a = S$ and $q = S + 1$, `reflectedTorusGrid_movesTo_coxeterPower`). The case $q = 1$ is not formalized. The formal proof follows Sections 5–9 with the following changes of bookkeeping.

(1) Lemma 5.2 allows any order among rows of equal height. The formalization fixes one order (`hybLtS`): at each integer height the arcs of later blocks lie lower, the arc of the last block leaving its last occurrence of the rank lies above all other arcs, and ι_r lies on top. With this order the hypotheses of Lemma 6.2 hold.

(2) After stage 3 of Lemma 7.1, and after the moves of Lemma 8.1, the arc o_r of the first block lies directly below the arc of the last block leaving its last occurrence of r. The column interval of the first arc lies strictly inside the column interval of the second, so one commutation exchanges the two rows (`LabeledGrid.exists_swapOuts`). These exchanges restore the fixed order and replace the appeal to Lemma 5.2.

(3) The formal block reduction (`exists_blockReduction`) leaves the columns in the order $B_{a-2}, \ldots, B_1, V, B_0$. In Lemma 7.1 the column V is first moved to the right past B_0. In Lemma 8.1 the O-stabilization is performed at V in this position, ν is moved to the bottom, κ is moved to the right past B_0, and then V is moved to the left past $B_{a-2}, \ldots, B_1$. Each of these exchanges is a commutation of columns whose segments are nested or disjoint.

(4) The admissible grid of β^q used in Lemma 8.1 is constructed with the cyclic matching of β^q computed explicitly (`betaPow_matching_one`, `betaPow_matching_two`), and it is related to $\texttt{wordGrid}(\beta^q)$ by `IsAdmissibleWordGrid.movesTo`.

11. Summary of dependencies

Ingredient	Source
Grid moves, `MovesTo`	`External/TauCeti/KnotTheory/Grid/Move/Basic.lean`
Destabilization from a 2×2 pattern	`isXStabilization_eraseOCorner`, `isDestabilization_swapMarkings`
Short segments	`columnsNoninterleaving_of_adjacent_rows`, `rowsNoninterleaving_of_adjacent_columns`
Admissible grids, same-height rows	`PositiveWordGridAdmissible.lean`, J10 chapter
$\beta \approx \delta$	`farRotationEquiv_coxeterWord` (J08)
Rotations and far commutations of word grids	Lemma 9.2, J09
Lemmas 3.2–3.4, 5.2–8.1	new, proved here

No step uses a geometric isotopy, an identification of grid diagrams with links, or a numerical computation. The chirality computation of Section 2 is evidence only.